

Image categorization: Pollen grains

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Problem Definition

The objective of this work is to develop an automated classification system capable of distinguishing between different types of pollen grains from microscopic images.

The classification problem involves training machine learning models to categorize grayscale images of pollen grains into one of 15 distinct classes. The dataset used in this study is sourced from the SYLVA data portal (its subset).

The goal is to explore different feature extraction methods (such as texture descriptors, histograms, and dimensionality reduction techniques) combined with various classification algorithms to determine the most effective approach for this multi-class classification problem.

Proposed Descriptors

This section describes the feature extractors implemented for the pollen grain classification task. A total of three feature extractors were developed, in addition to the baseline gray levels extractor provided as a template. Each extractor captures different aspects of the pollen grain images, ranging from simple pixel-level features to sophisticated texture and shape descriptors.

LBP Histogram Feature Extractor

The Local Binary Pattern (LBP) Histogram Feature Extractor captures texture information by analyzing local patterns in pixel neighborhoods:

Method

- Computes LBP codes for each pixel by comparing its intensity with neighboring pixels arranged in a circular pattern
- Each neighbor pixel is assigned a binary value (1 if intensity $\geq$ center pixel, 0 otherwise)
- The binary pattern is converted to a decimal LBP code
- A histogram of LBP codes is computed, optionally using a spatial grid division

Parameters

- radius: Radius of the circular neighborhood (default: 1.0)
- neighbors: Number of sampling points in the circular neighborhood (default: 8)
- grid_rows: Number of rows for spatial histogram division (default: 1, no division)
- grid_cols: Number of columns for spatial histogram division (default: 1, no division)

Feature Vector

- Without grid: Single histogram of size $2^{\text{neighbors}}$ bins (e.g., 256 bins for 8 neighbors)
- With grid: Concatenated histograms from each grid cell, normalized using L1 norm
- The histogram is normalized to sum to 1.0, making it scale-invariant

Advantages:

- Captures local texture patterns effectively
- Robust to monotonic illumination changes
- Computationally efficient
- Spatial grid division can preserve some spatial information while maintaining compactness

HOG Feature Extractor

The Histogram of Oriented Gradients (HOG) Feature Extractor describes the shape and appearance of objects by analyzing the distribution of gradient orientations

Method

- Divides the image into cells and computes histograms of gradient orientations within each cell
- Groups cells into overlapping blocks for normalization
- Uses OpenCV's HOGDescriptor implementation
- The descriptor captures edge and contour information, which is crucial for distinguishing pollen grain shapes

Parameters

- win_width: Width of the detection window (default: image width)

- win_height: Height of the detection window (default: image height)
- block_size: Size of blocks in pixels (default: 16)
- block_stride: Stride between blocks in pixels (default: 8)
- cell_size: Size of cells in pixels (default: 8)
- nbins: Number of orientation bins in the histogram (default: 9)

Feature Vector

- The HOG descriptor produces a feature vector whose length depends on the window size and block/cell configuration
- For a 64×64 image with default parameters, this typically results in several hundred features
- Features are automatically normalized by the HOG descriptor

Advantages

- Excellent for capturing shape and edge information
- Invariant to geometric and photometric transformations
- Well-established descriptor with proven effectiveness in object recognition tasks
- Combined LBP+HOG Feature Extractor

The Combined LBP+HOG Feature Extractor

Integrates both texture and shape information by concatenating features from both descriptors

Method

- Applies histogram equalization preprocessing to enhance image contrast
- Extracts LBP histogram features using the LbpExtractor
- Extracts HOG features using the HogExtractor
- Normalizes each feature vector independently using L2 normalization
- Concatenates the normalized LBP and HOG features into a single combined feature vector

Parameters

- Accepts all parameters from both LBP and HOG extractors:

- LBP: lbp_radius, lbp_neighbors, lbp_grid_rows, lbp_grid_cols
- HOG: hog_win_width, hog_win_height, hog_block_size, hog_block_stride, hog_cell_size, hog_nbins
- Default values match those of the individual extractors

Feature Vector

- Concatenation of normalized LBP and HOG feature vectors
- The dimensionality is the sum of LBP and HOG feature dimensions
- L2 normalization ensures both feature types contribute equally to the combined representation

Advantages:

- Combines complementary information: texture (LBP) and shape (HOG)
- Histogram equalization preprocessing improves feature quality
- Normalization prevents one feature type from dominating the other
- Potentially captures more discriminative information than individual descriptors alone

PCA Dimensionality Reduction

Principal Component Analysis (PCA) is implemented as an optional post-processing step that can be applied to features extracted by any of the feature extractors described above. PCA serves as a dimensionality reduction technique that helps reduce overfitting and computational complexity while preserving the most important information in the feature space.

Method

PCA is applied after feature extraction, transforming the original high-dimensional feature vectors into a lower-dimensional space. The process involves:

- Computing the covariance matrix of the training features
- Finding the principal components that capture the maximum variance in the data

- Selecting the number of components based on either a variance threshold or a fixed number
- Projecting both training and validation features onto the reduced-dimensional space using the principal components learned from the training set only

Parameters

- `variance_threshold`: Fraction of variance to retain (default: 0.85, meaning 85% of variance is preserved)
- `max_components`: Maximum number of principal components to use (default: 0, which means use variance threshold instead)

Feature Vector

The dimensionality of the output features depends on the selected number of components

- If using variance threshold: The number of components is automatically determined to retain the specified percentage of variance
- If using `max_components`: The feature vector is reduced to exactly that many dimensions

The transformation preserves the most discriminative information while significantly reducing dimensionality

Advantages

- Reduces overfitting by eliminating noise and redundant information
- Decreases computational cost for training and prediction
- Improves generalization by focusing on the most important feature directions
- Can be applied to any feature extractor (Gray Levels, LBP, HOG, or LBP+HOG)
- The PCA model is trained only on the training set and then applied to validation/test sets, ensuring proper cross-validation

Implementation Details

- PCA is implemented using OpenCV's `cv::PCA` class
- The PCA model (eigenvectors, eigenvalues, and mean) can be saved and loaded for later use

- During training, PCA is fitted on the training features and then transforms both training and validation features
- During testing, the same PCA transformation learned during training is applied to test features
- This ensures consistency between training and testing phases

Usage

PCA can be optionally enabled when training classifiers. When enabled, it is applied after feature extraction but before classifier training. This makes it particularly useful for high-dimensional features such as Gray Levels (4096 dimensions for 64×64 images) or combined LBP+HOG features, where dimensionality reduction can significantly improve both performance and efficiency.

Experiments and Results

This section presents a comprehensive analysis of the experiments conducted using three different classifiers (K-NN, SVM, and RTrees) with various feature extractors (LBP, HOG, and their combination LBP+HOG). All experiments were performed on a dataset containing 24,000 training samples and 3,000 validation samples, representing 15 pollen classes.

The experiments were designed to evaluate the performance of different classifier-feature extractor combinations. The main hyperparameters explored include:

- K-NN: Parameter K ranging from 1 to 11
- SVM: Kernel types (Linear, RBF, CHI2), regularization parameter C (0.01 to 100.0), and gamma parameter (0.0001 to 10.0)
- RTrees: Number of features per node V (0, 10, 20, 30), maximum number of trees T (50, 100, 200, 500), and OOB error threshold E (0.0001, 0.001, 0.01)

Feature extractor parameters were also varied, including LBP radius (0.5-2.0), neighbors (4-8-18), grid configurations (1×1 to 5×5), and HOG block sizes, strides, and cell sizes optimized for feature dimensionality reduction.

Classifier Comparison

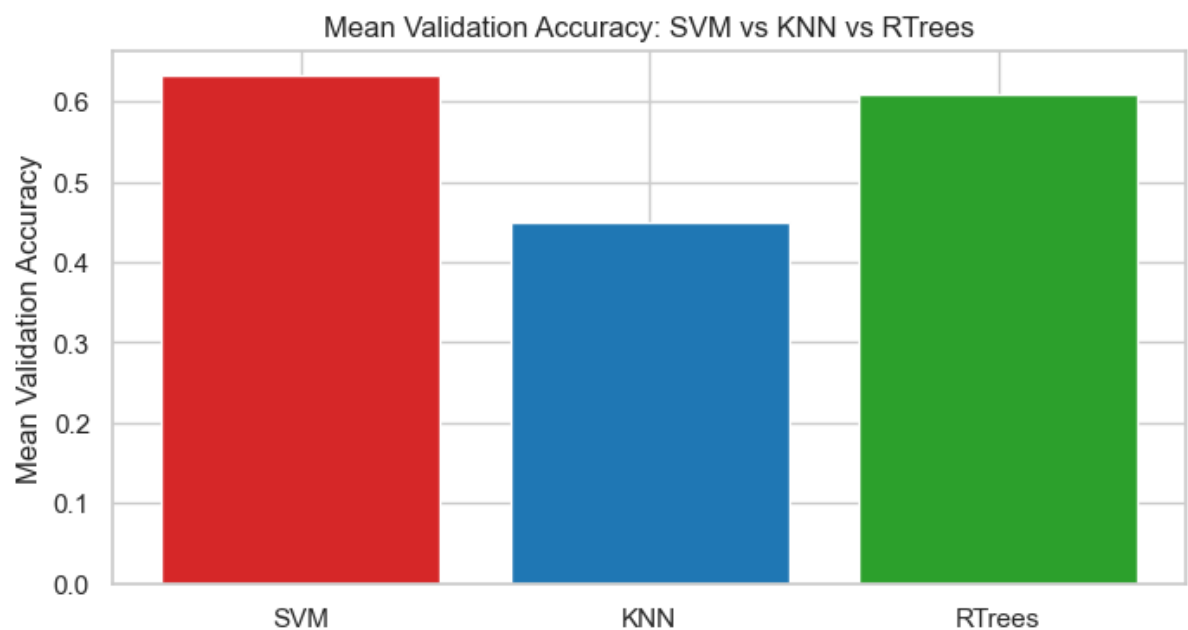


Figure 1: Mean validation accuracy comparison across the three classifiers tested. The bar chart shows the average performance of SVM, KNN, and RTrees across all hyperparameter configurations.

The mean validation accuracy comparison reveals that SVM achieves the highest average performance among the three classifiers, followed by RTrees and KNN. This suggests that SVM's ability to find optimal decision boundaries in high-dimensional feature spaces is particularly effective for pollen classification.

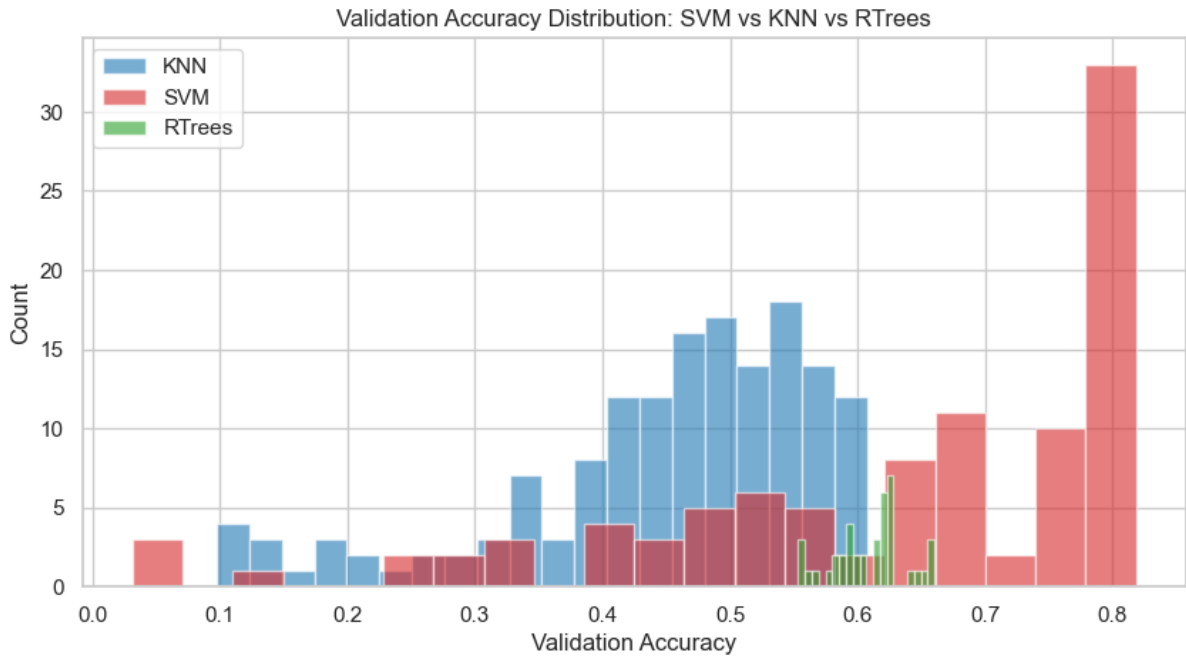


Figure 2: Distribution of validation accuracy values across all experiments for each classifier. The histogram shows the spread and variability of performance for SVM, KNN, and RTrees

The distribution analysis shows that SVM experiments exhibit a wider range of validation accuracies, indicating that hyperparameter selection is crucial for this classifier. RTrees shows more consistent performance across configurations, while KNN demonstrates moderate variability.

K-NN Classifier Analysis

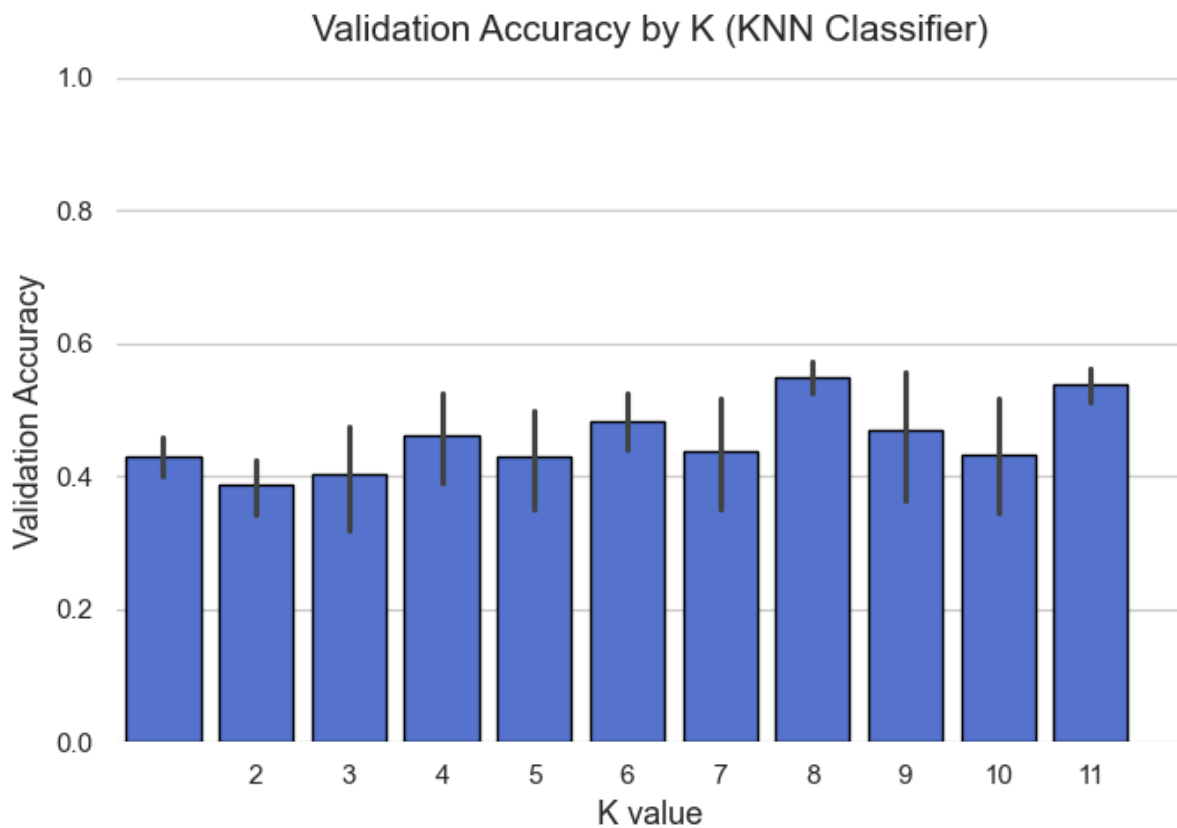


Figure 3: Validation accuracy as a function of the K parameter for K-NN classifier. The bar chart shows how different values of K affect classification performance.

The analysis of KNN performance with respect to parameter K reveals that optimal performance is achieved at intermediate K values. The best K values typically fall in the range of 3-7, balancing between local detail and generalization.

SVM Classifier Analysis

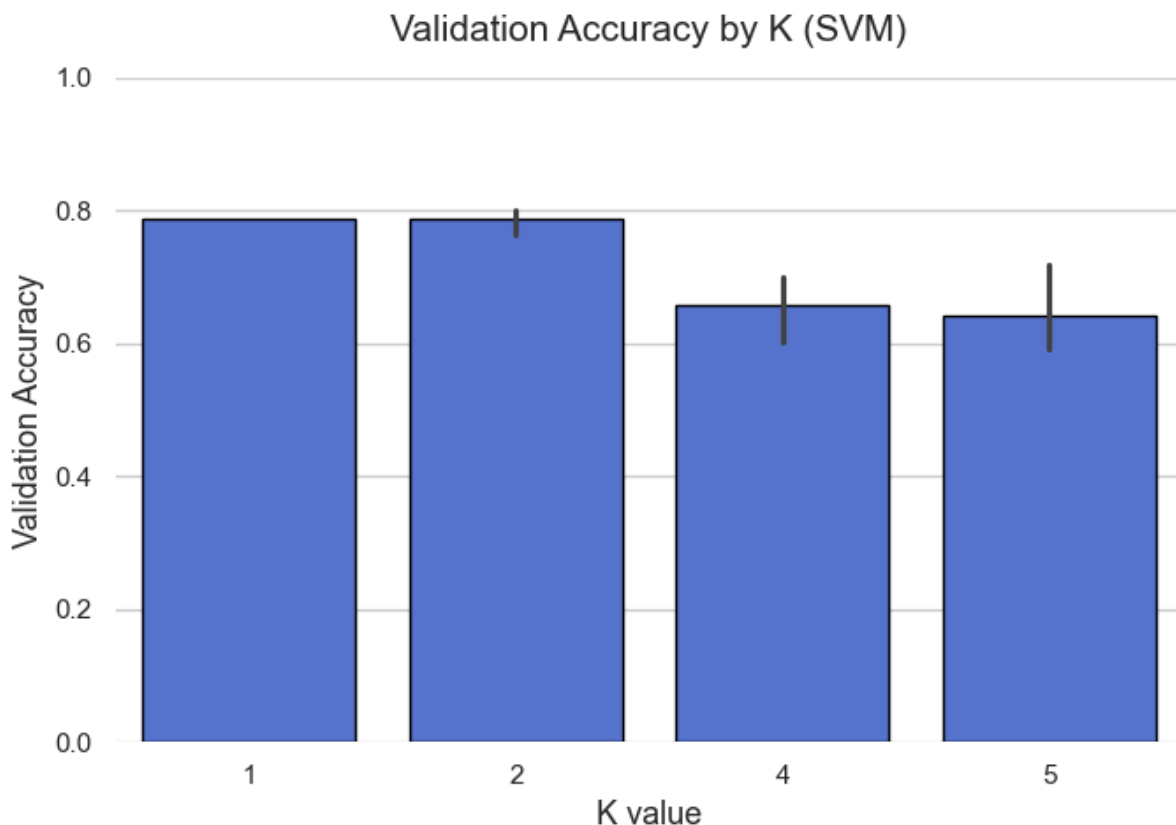


Figure 4: Validation accuracy as a function of SVM kernel type (K). Kernel types include polynomial (K=1), RBF (K=2), and CHI2 (K=4) and INTER(K=5). The chart shows performance differences between kernel types.

The SVM kernel analysis (Figure 4) demonstrates that different kernel types exhibit distinct performance characteristics. The CHI2 kernel (K=4) shows particular promise for histogram-based features like LBP, as it is specifically designed for comparing probability distributions but it is performing bad with PCA reduced dimensions. The RBF kernel (K=2) provides good general performance (best among all classifiers), while the polynomial kernel (K=1) may have same performance as RBF but training time was longer compared to rbf and CHI2.

Hyperparameter tuning experiments revealed that C parameter's Lower values (0.1-1.5) generally provide better generalization, reducing overfitting. Gamma parameter: For RBF and CHI2 kernels, moderate gamma values (0.01-1.0) balance between model complexity and generalization, but for the final model little higher gamma was chosen but with strong overfitting to training data (accuracy gain from that was around 1-1.5%).

Feature extractor combination: LBP+HOG features consistently outperform individual LBP or HOG features when used with SVM

Confusion Matrix Analysis

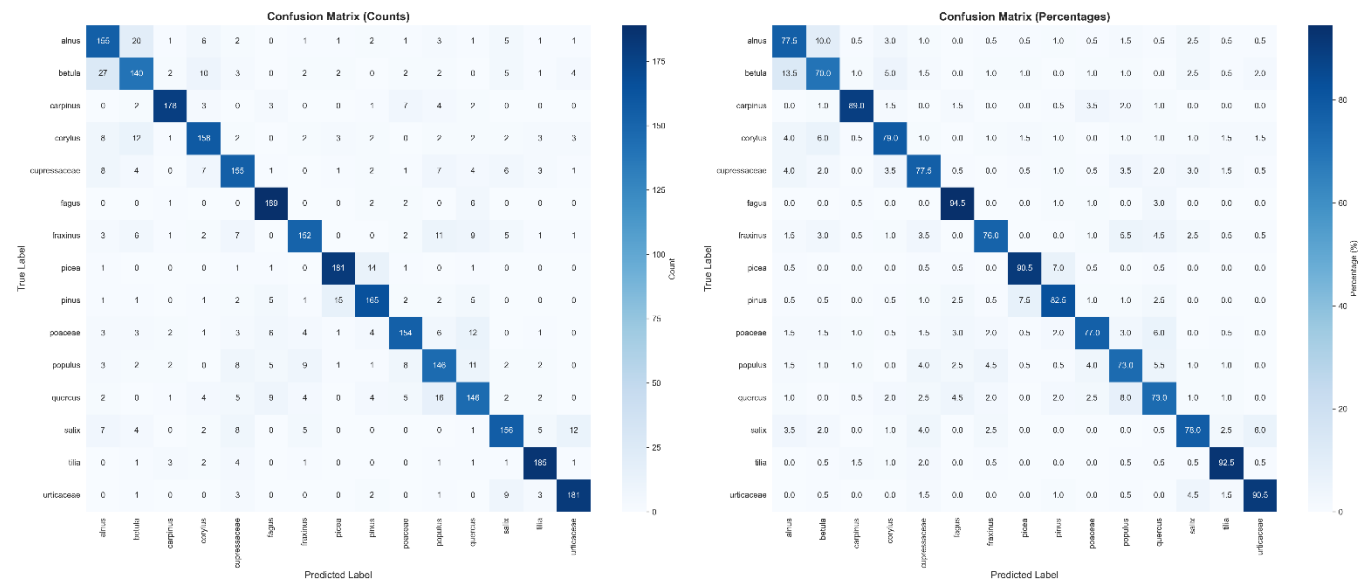


Figure 5: Confusion matrix for the best-performing model on the validation set. The matrix shows which pollen classes are frequently confused with each other. Diagonal elements represent correct classifications, while off-diagonal elements indicate misclassifications.

The confusion matrix analysis provides insights into class-wise performance and common misclassification patterns. Some pollen classes show high confusion rates between them (alnus and Betula for example), likely due to:

1. Visual similarity, certain pollen types may have similar morphological characteristics
2. Feature representation limitations: The chosen features may not capture distinguishing characteristics for certain class pairs (combination of only two descriptors was used)

The diagonal elements represent correct classifications, and the analysis reveals that most classes achieve good recognition rates. However, as mentioned earlier specific class pairs show higher confusion, suggesting areas for future improvement through feature engineering or data augmentation, or retraining on these specific areas

Key Findings

1. **Best Performing Configuration:** The combination of LBP+HOG features with SVM classifier using RBF kernel ($K=2$) and optimized C and gamma parameters achieves the highest validation accuracy
2. **Feature Extractor Impact:** The combined LBP+HOG feature extractor consistently outperforms individual LBP or HOG extractors, demonstrating the complementary nature of texture (LBP) and shape (HOG) features
3. **Hyperparameter Sensitivity:** SVM performance is highly sensitive to hyperparameter selection, particularly the C and gamma parameters. Systematic grid search is essential for optimal performance
4. **Dimensionality Reduction:** The use of PCA with variance thresholds around 85-95% helps reduce overfitting while maintaining classification performance, particularly important for high-dimensional feature spaces, but most importantly it reduces model size (can be change twice or thrice in size).
5. **Model Size Considerations:** While larger feature spaces (e.g., 5×5 LBP grids) may provide more discriminative power, they also increase model size. A balance must be struck between accuracy and model size constraints.

Performance Metrics Summary

The experiments demonstrate that:

- SVM RBF with LBP+HOG achieves the best overall validation accuracy
- RTrees provides competitive performance with better consistency across configurations
- K-NN offers simplicity but lower peak performance compared to the other classifiers
- PCA is very good for retaining performance while reducing dimensionality and model size

Conclusions

This work presents a comprehensive evaluation of automated pollen grain classification using various feature extraction methods and machine learning classifiers. Through systematic experimentation with LBP, HOG, and combined LBP+HOG feature extractors, combined with K-NN, SVM, and RTrees classifiers, several key insights have been established.

The best-performing configuration identified in this study is the SVM classifier with RBF kernel ($K=2$ $C=2.0$ $G=5.5$) using combined LBP+HOG features (with parameters: “1.5 6 5 5 64 64 32 16 16 9” and PCA 0.88) , which achieved the highest validation accuracy among all tested configurations. combination of these features captures both local texture patterns and global shape characteristics, providing a more discriminative representation than either descriptor alone.

The experimental results reveal that SVM consistently outperforms both RTrees and KNN across different hyperparameter configurations. However, this superior performance comes with increased sensitivity to hyperparameter selection, particularly C and γ . grid search approach proved essential for achieving optimal performance, with lower C values (0.1-2.0) generally providing better generalization.

PCA successfully reduced model size while maintaining classification performance. By retaining 85-95% of variance, PCA helps fight overfitting and computational complexity, making it particularly beneficial for combined LBP+HOG descriptor.

The confusion matrix analysis revealed that while most pollen classes achieve good recognition rates, certain class pairs (such as *alnus* and *betula*) show higher confusion rates. This suggests that while the chosen feature extractors are effective overall, there may be limitations in distinguishing visually similar pollen types. Future work could address this through more sophisticated feature engineering, data augmentation techniques, or ensemble methods.

Overall, this study successfully demonstrates that traditional computer vision techniques, when properly combined and optimized, can achieve strong performance (validation accuracy above 80%) for automated pollen grain classification.